

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T53 · Statistics

With Answers

Questions with full worked solutions

■ 84 pages

FAMILY MAP

Statistics Toolkit

F01	Estimate grouped-data measures	14 Q 44 marks
F02	Calculate and reverse discrete averages	16 Q 51 marks
F03	Measure and compare spread	6 Q 14 marks
F04	Present and interpret statistical diagrams	3 Q 8 marks

- 1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean distance.

..... km
(4)

Worked Solution 01

Grouped data: modal class and estimated mean

UNIT 2**5 marks****Modal class**

The greatest frequency is 40, so $5 < d \leq 10$.

Estimated mean

$$\bar{d} = \frac{2.5(26) + 7.5(40) + 12.5(16) + 17.5(10) + 22.5(8)}{100} = \frac{920}{100} = 9.2.$$

Final answer

$$5 < d \leq 10, \quad \bar{d} = 9.2 \text{ km}$$

- 1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14
--	--	---	--	----	----

Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

Worked Solution 16

Ordered cards: range, median and mode

UNIT 2
3 marks**Complete the ordered list**

The list is 4, 6, 6, 9, 10, 14.

Check the conditions

$$14 - 4 = 10, \quad \frac{6 + 9}{2} = 7.5, \quad \text{mode} = 6.$$

Final answer

4, 6, 9 in the three hidden positions

Question 12

4MA1-2H Q1

ORIGINAL QUESTION

3 marks

- 1 Here are eight numbers written in order of size

$$h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

 $h = \dots\dots\dots$ $j = \dots\dots\dots$ $k = \dots\dots\dots$

Worked Solution 12

Median, mode and range

MODULAR UNIT 2

3 marks

Median

$$(8 + j)/2 = 10 \Rightarrow j = 12.$$

Mode and range

$$k = 18, \quad k - h = 13 \Rightarrow h = 5.$$

Final answer

$$h = 5, \quad j = 12, \quad k = 18$$

Question 17

4MA1-2H Q12

ORIGINAL QUESTION

3 marks

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800

For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

Worked Solution 17

Combined mean

MODULAR UNIT 2

3 marks

Totals

$$4(11800) = 47200, \quad 3(13207) = 39621.$$

Mean

$$(47200 + 39621)/7 = 12403.$$

Final answer

12403 steps

Question 21

4MA1-2H Q1 (a)

ORIGINAL QUESTION UNIT 2

1 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (a) Write down the modal class.

.....
(1)

Worked solution 21

Modal class

MODULAR UNIT 2

1 marks

(a) – Modal class

Read the greatest frequency

The greatest frequency is 18, in the class $25 < m \leq 30$.**Final answer** $25 < m \leq 30$

Question 22

4MA1-2H Q1 (b)

ORIGINAL QUESTION UNIT 2

3 marks

- 1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	Frequency
$0 < m \leq 5$	8
$5 < m \leq 10$	2
$10 < m \leq 15$	6
$15 < m \leq 20$	4
$20 < m \leq 25$	12
$25 < m \leq 30$	18

- (b) Work out an estimate for the total length, in minutes, of these telephone calls.

..... minutes
(3)

Worked solution 22

Estimated total length from grouped data

MODULAR UNIT 2

3 marks

(b) – Estimated total length from grouped data

Use class midpoints

$$2.5(8) + 7.5(2) + 12.5(6) + 17.5(4) + 22.5(12) + 27.5(18).$$

Add the midpoint products

$$20 + 15 + 75 + 70 + 270 + 495 = 945.$$

Final answer

945 minutes

Question 27

4MA1-2H Q8 whole

ORIGINAL QUESTION UNIT 2

3 marks

- 8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

$k = \dots\dots\dots$

Worked solution 27

Mean after two further matches

MODULAR UNIT 2

3 marks

whole – Mean after two further matches

Find the known totals

$$\text{First total} = 8(6) = 48, \quad \text{new total} = 10(7) = 70.$$

Find the two new goals

$$2k = 70 - 48 = 22.$$

Solve

$$k = 11.$$

Final answer

$$k = 11$$

Question 28

4MA1-2H Q13 whole

ORIGINAL QUESTION UNIT 2

2 marks

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

.....

Worked solution 28

Interquartile range from eleven values

MODULAR UNIT 2

2 marks

whole – Interquartile range from eleven valuesRead Q_1 and Q_3

$$Q_1 = 4 \text{ (middle of the lower five),} \quad Q_3 = 13 \text{ (middle of the upper five).}$$

Subtract

$$\text{IQR} = Q_3 - Q_1 = 13 - 4 = 9.$$

Final answer

9

Question 02

4MA1-1H Q2

ORIGINAL QUESTION

3 marks

- 2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	Frequency
$0 < M \leq 30$	5
$30 < M \leq 60$	6
$60 < M \leq 90$	8
$90 < M \leq 120$	9
$120 < M \leq 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

..... minutes

Worked Solution 02

Estimated grouped total

MODULAR UNIT 2

3 marks

Use class midpoints $15, 45, 75, 105, 135.$ **Weight and add**

$$15(5) + 45(6) + 75(8) + 105(9) + 135(2) = 2160.$$

Final answer

2160 minutes

Question 10

4MA1-1H Q10

ORIGINAL QUESTION

2 marks

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

Worked Solution 10

Interquartile range

MODULAR UNIT 2

2 marks

Locate the quartiles

$$Q_1 = 15, \quad Q_3 = 25.$$

Subtract

$$\text{IQR} = 25 - 15 = 10.$$

Final answer

10

Question 23

4MA1-2H Q10

ORIGINAL QUESTION

4 marks

10 Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17

After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

Worked Solution 23

Difference between two final-round scores

MODULAR UNIT 2

4 marks

Find the earlier totals

$$A : 11(17) = 187, \quad B : 9(18) = 162.$$

Find the final totals

$$A : 12(18.5) = 222, \quad B : 10(18.5) = 185.$$

Compare the last rounds

$$A \text{ scored } 222 - 187 = 35, \quad B \text{ scored } 185 - 162 = 23; \quad 35 - 23 = 12.$$

Final answer*12points*

Original question 01

4MA1-1H Q1 M01

MODULAR UNIT 2

1 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (a) Write down the modal class.

.....
(1)

Worked solution 01

Identify the modal class

MODULAR UNIT 2

1 marks

M01 – Identify the modal class**Read the largest frequency**

The largest frequency is 18, in the interval $10 < p \leq 15$.

Final answer

$$10 < p \leq 15$$

Original question 02

4MA1-1H Q1 M02

MODULAR UNIT 2

4 marks

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

Worked solution 02

Estimate the grouped mean

MODULAR UNIT 2

4 marks

M02 – Estimate the grouped mean**Use class midpoints**

$$12.5(18) + 17.5(16) + 22.5(14) + 27.5(8) + 32.5(4) = 1170.$$

Divide by the total frequency

$$\bar{p} = 1170/60 = 19.5.$$

Final answer

19.5 dollars

Original question 22

4MA1-2H Q8 M01

MODULAR UNIT 2

3 marks

- 8 Kim puts 4 bananas in a bag.
The mean weight of the 4 bananas in the bag is 145 grams.

Andy puts one more banana into the bag.
The mean weight of the 5 bananas in the bag is 142 grams.

Work out the weight of the banana that Andy puts into the bag.

..... grams

Worked solution 22

Use two means to find a missing weight

MODULAR UNIT 2

3 marks

M01 – Use two means to find a missing weight**Find the old total**

$$4(145) = 580 \text{ g.}$$

Find the new total

$$5(142) = 710 \text{ g.}$$

Subtract

$$710 - 580 = 130 \text{ g.}$$

Final answer

130 g

Original question 24

4MA1-2H Q10 M01

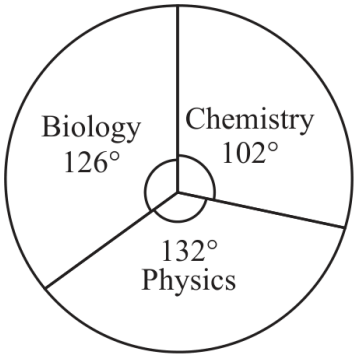
MODULAR UNIT 2

5 marks

- 10 All the students in year 10 and all the students in year 11 named their favourite science subject from Biology, Chemistry and Physics.

The pie chart shows information about the results for the year 10 students.

The table shows information about the results for the year 11 students.



Pie chart for year 10

science subject	number of students
Biology	$3x + 6$
Chemistry	$5x + 8$
Physics	$7x - 9$

Table for year 11

There are 300 students in year 10

There are 320 students in year 11

More students in year 10 than in year 11 said Biology was their favourite science subject.

How many more?

Original question 24 (continued)

4MA1-2H Q10 M01

MODULAR UNIT 2

5 marks

.....

Worked solution 24

Compare Biology counts from a pie chart and table

MODULAR UNIT 2

5 marks

M01 – Compare Biology counts from a pie chart and table**Solve the Year 11 total**

$$(3x + 6) + (5x + 8) + (7x - 9) = 320 \Rightarrow 15x + 5 = 320 \Rightarrow x = 21.$$

Find Year 11 Biology

$$3x + 6 = 69.$$

Read Year 10 Biology from the pie chart

$$(126/360) \times 300 = 105.$$

Compare

69-105=-36, so the number differs by 36 students.

Final answer

36 students

Original question 20

4MA1-2H Q1 Q1

ORIGINAL QUESTION**3 marks**

- 1 Here are some integers where $a < b < c < d$

$a \qquad b \qquad c \qquad d \qquad d \qquad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

$d = \dots\dots\dots$

Worked solution 20

Chapter: Statistics Toolkit

MODULAR UNIT 2

3 marks

Q1 – Chapter: Statistics Toolkit

Family: Calculate and reverse discrete averages • Variant: Construct a data set from stated averages and range

Method

Because d occurs three times, the mode gives $d = 9$.

Why: The repeated largest entry is the unique modal value.

Method

The median gives $(c + d)/2 = 8$, so $c = 7$.

Why: There are six ordered values, so the median is the mean of the third and fourth.

Method

The range gives $d - a = 4$, so $a = 5$; the only integer strictly between 5 and 7 is $b = 6$.

Why: Use the range and the strict integer ordering.

Final answer

a: $a = 5$

Final answer

b: $b = 6$

Final answer

c: $c = 7$

Final answer

d: $d = 9$

Original question 25

4MA1-2H Q12 Q12

ORIGINAL QUESTION

3 marks

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

 $W = \dots\dots\dots$

Worked solution 25

Chapter: Statistics Toolkit

MODULAR UNIT 2

3 marks

Q12 – Chapter: Statistics Toolkit

Family: Calculate and reverse discrete averages • Variant: Combine or separate subgroup means

MethodThe total mass of all seven parcels is $7 \times 2.7 = 18.9$ kg.*Why: Mean multiplied by count gives the total.***Method**The other four parcels total $4 \times 3.3 = 13.2$ kg, so the three equal parcels total $18.9 - 13.2 = 5.7$ kg.*Why: Subtract the known group total.***Method**Therefore $W = 5.7/3 = 1.9$ kg.*Why: Divide the remaining total equally among three parcels.***Final answer****W:** $W = 1.9$ kg

Original question 04

4MA1-1H Q5 Q5(a)

ORIGINAL QUESTION**3 marks**

5 Jenny has six cards.

Each card has a whole number written on it so that

the smallest number is 5

the largest number is 24

the median of the six numbers is 14

the mode of the six numbers is 8

Jenny arranges her cards so that the numbers are in order of size.

5					24
---	-------	-------	-------	-------	----

(a) For the remaining four cards, write on each dotted line a number that could be on the card.

(3)

Worked solution 04

Chapter: Statistics Toolkit

MODULAR UNIT 2

3 marks

Q5(a) – Chapter: Statistics Toolkit

Family: Calculate and reverse discrete averages • Variant: Construct a data set from stated averages and range

Method

A valid ordered set is 5, 8, 8, 20, 21, 24.

Why: It respects the smallest and largest cards.

Method

The middle pair is 8 and 20, whose mean is $(8+20)/2 = 14$.

Why: The median of six ordered values is the mean of the third and fourth.

Method

The mode is 8, because 8 occurs twice and the other listed values occur once.

Why: Check the stated mode.

Final answer

six cards: 5, 8, 8, 20, 21, 24

Original question 05

4MA1-1H Q5 Q5(b)

ORIGINAL QUESTION**3 marks**

A basketball team plays 6 games.

After playing 5 games, the team has a mean score of 21 points per game.

After playing 6 games, the team has a mean score of 23 points per game.

(b) Work out the number of points the team scored in its 6th game.

(3)

Worked solution 05

Chapter: Statistics Toolkit

MODULAR UNIT 2

3 marks

Q5(b) – Chapter: Statistics Toolkit

Family: Calculate and reverse discrete averages • Variant: Find a required score for a target mean

MethodAfter five games the total score is $5 \times 21 = 105$.*Why: Mean multiplied by number of games gives the total.***Method**After six games the total score is $6 \times 23 = 138$.*Why: Use the second mean.***Method**Sixth-game score = $138 - 105 = 33$ points.*Why: Subtract the first five-game total.***Final answer****sixth-game score: 33 points**

Original question 23

4MA1-2H Q9

ORIGINAL QUESTION**4 marks**

- 9 The frequency table gives information about the number of points scored by a player.

Number of points	Frequency
0	13
1	17
2	8
3	x
4	11

The mean number of points scored is 2

Work out the value of x

$x =$

Worked solution 23

Chapter: Statistics Toolkit

MODULAR UNIT 2

4 marks

Q9 – Chapter: Statistics Toolkit

Family: Calculate and reverse discrete averages • Variant: Recover a missing value or frequency from a mean

Method

The weighted total is $0 \times 13 + 1 \times 17 + 2 \times 8 + 3x + 4 \times 11 = 77 + 3x$.

Why: Multiply each value by its frequency.

Method

The total frequency is $13 + 17 + 8 + x + 11 = 49 + x$.

Why: Add all frequencies.

Method

$(77 + 3x) / (49 + x) = 2$, so $77 + 3x = 98 + 2x$.

Why: Use the stated mean.

Method

$x = 21$.

Why: Solve the linear equation.

Final answer

x: 21

Original question 28

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

- 15 Diyar recorded the distance, in kilometres, that he cycled each day for 11 days.
Here are his results.

8 10 12 13 5 23 21 7 5 16 14

Find the interquartile range of his results.

..... km

Worked solution 28

Chapter: Statistics Toolkit

MODULAR UNIT 2

3 marks

Q15 – Chapter: Statistics Toolkit

Family: Measure and compare spread • Variant: Find quartiles and IQR from discrete data

Method

Ordered data: 5,5,7,8,10,12,13,14,16,21,23.

*Why: Sort the 11 distances.***Method**

The lower and upper quartiles are 7 and 16.

*Why: Use the quartile positions for 11 values.***Method** $\text{IQR} = 16 - 7 = 9 \text{ km.}$ *Why: Subtract the quartiles.***Final answer****IQR: 9 km**

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

- (a) Write down the modal class.

(1)

Worked solution

January 2020 | Question 4 | 1 marks

Family: Estimate grouped-data measures • Variant: Identify the modal class

Method / working

1. The largest frequency is 16 in the $3 < w \leq 4$ class.

Final answer: $3 < w \leq 4$ kg

- 4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w \leq 3$	12
$3 < w \leq 4$	16
$4 < w \leq 5$	9
$5 < w \leq 6$	2
$6 < w \leq 7$	1

(a) Write down the modal class.

(b) Work out an estimate for the mean weight of the 40 babies.

..... kg
(4)

Worked solution

January 2020 | Question 4 | 4 marks

Family: Estimate grouped-data measures • Variant: Estimate the mean from grouped intervals

Method / working

1. Use class midpoints 2.5, 3.5, 4.5, 5.5, 6.5.
 2. Weighted total = $30 + 56 + 40.5 + 11 + 6.5 = 144$.
 3. Estimated mean = $144 / 40$.
 4. Mean = 3.6 kg.
- Final answer: 3.6 kg

- 2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	Frequency
$40 < s \leq 50$	10
$50 < s \leq 60$	16
$60 < s \leq 70$	19
$70 < s \leq 80$	23
$80 < s \leq 90$	12

- (a) Write down the modal class.

(1)

Worked solution

January 2021 | Question 2 | 1 marks

Family: Estimate grouped-data measures • Variant: Identify the modal class

Method / working

1. The largest frequency is 23 in the $70 < s \leq 80$ class.

Final answer: $70 < s \leq 80$

- (b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge.
Give your answer correct to 3 significant figures.

..... kilometres per hour
(4)

Worked solution

January 2021 | Question 2 | 4 marks

Family: Estimate grouped-data measures • Variant: Estimate the mean from grouped intervals

Method / working

1. Use midpoints 45,55,65,75,85 with frequencies 10,16,19,23,12.
2. Weighted total = 5310.
3. Mean estimate = $5310/80 = 66.375$.
4. To 3 s.f., the mean is 66.4 km/h.

Final answer: 66.4 km/h

2 Ava writes down five whole numbers.

For these five numbers

the median is 7

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

.....

Worked solution

January 2021 | Question 2 | 3 marks

Family: Calculate and reverse discrete averages • Variant: Construct a data set from stated averages and range

Method / working

1. Use an ordered list with median 7 and mode 8.
2. Choose 4 as the smallest value so the range is 5.
3. The list 3,7,8,8,4 reordered as 3,4,7,8,8 satisfies all conditions.

Final answer: 3, 7, 8, 8, 4

- 7 There are 5 cocoa pods in a bag.
The mean weight of the 5 cocoa pods is 398 grams.

A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.

Work out the weight of the sixth cocoa pod that is put into the bag.

..... grams

Worked solution

January 2021 | Question 7 | 3 marks

Family: Calculate and reverse discrete averages • Variant: Recover a missing value or frequency from a mean

Method / working

1. Five-pod total = $5 \times 398 = 1990$ g.

2. Six-pod total = $6 \times 401 = 2406$ g.

3. Sixth pod = $2406 - 1990 = 416$ g.

Final answer: 416 g

12 15 people were asked how long, in minutes, they had been waiting for a bus.

Here are the results.

2 3 3 4 5 6 6 8 9 10 11 13 14 15 18

Find the interquartile range of these times.

..... minutes

Worked solution

January 2021 | Question 12 | 2 marks

Family: Measure and compare spread • Variant: Find quartiles and IQR from discrete data

Method / working

1. For 15 ordered values, Q1 is the 4th value (4) and Q3 is the 12th value (13).
2. $IQR = 13 - 4 = 9$ minutes.

Final answer: 9 minutes

10 Jethro has sat 5 tests.

Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

Worked solution

June 2021 | Question 10 | 3 marks

Family: Calculate and reverse discrete averages • Variant: Find a required score for a target mean

Method / working

1. The first five tests total $5 \times 74 = 370$.
 2. A mean of 77 after six tests would total $6 \times 77 = 462$.
 3. The sixth mark is $462 - 370 = 92$.
- Final answer: 92

14 Here is the number of goals that Henri's team scored one summer in each water polo match.

5 8 9 11 13 13 14 15 16 17 20

Find the interquartile range of the numbers of goals.
Show your working clearly.

Worked solution

June 2021 | Question 14 | 2 marks

Family: Measure and compare spread • Variant: Find quartiles and IQR from discrete data

Method / working

1. Identify the lower quartile 9 and upper quartile 16 from the ordered data.
2. $IQR = 16 - 9 = 7$.

Final answer: 7

- 3 The table gives information about the amounts of money, in euros, that 70 of Anjali's friends spent last Saturday.

Money spent (S euros)	Frequency
$0 < S \leq 8$	6
$8 < S \leq 16$	14
$16 < S \leq 24$	19
$24 < S \leq 32$	25
$32 < S \leq 40$	6

One of Anjali's 70 friends is going to be chosen at random.

- (a) Find the probability that this friend spent more than 24 euros last Saturday.

.....
(1)

Worked solution

November 2021 | Question 3 | 1 marks

Family: Present and interpret statistical diagrams | Variant: Read values and compare categories from a basic diagram

METHOD / WORKING

1. More than 24 euros:
2. $25+6=31$
3. Total frequency:
4. $6+14+19+25+6=70$
5. $P(S > 24) = \frac{31}{70}$
6. Use midpoints 4, 12, 20, 28, 36.
7. $4(6)+12(14)+20(19)+28(25)+36(6)=1488$
8. estimated mean $= \frac{1488}{70} = 21.257 \dots$
9. Answer: $\frac{31}{70}$, estimated mean 21.3 euros.

FINAL ANSWER

 $\frac{31}{70}$, estimated mean 21.3 euros.

- (b) Work out an estimate for the mean amount of money spent by Anjali's friends last Saturday.
Give your answer correct to 2 decimal places.

..... euros
(4)

Worked solution

November 2021 | Question 3 | 4 marks

Family: Estimate grouped-data measures | Variant: Estimate the mean from grouped intervals

METHOD / WORKING

1. More than 24 euros:

2. $25+6=31$

3. Total frequency:

4. $6+14+19+25+6=70$ 5. $P(S > 24) = \frac{31}{70}$

6. Use midpoints 4, 12, 20, 28, 36.

7. $4(6)+12(14)+20(19)+28(25)+36(6)=1488$ 8. estimated mean $= \frac{1488}{70} = 21.257 \dots$ 9. Answer: $\frac{31}{70}$, estimated mean 21.3 euros.

FINAL ANSWER

 $\frac{31}{70}$, estimated mean 21.3 euros.

- 4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

Answer reference | May-Nov 2020 | Question 4

Family: Calculate and reverse discrete averages • Variant: Calculate the mean from a list or discrete frequency table

4	$\frac{x+10}{2} = 9$ or $x = 8$		4	M1 (indep)
	$\frac{4+7+x+10+y+y}{6} = 11$ oe or '66' - 4 - 7 - 10 (= 45)			M1 where x may be a number $7 < x < 10$
	$(y =) (6 \times 11 - 4 - 7 - 10 - '8') \div 2$			M1 ft their median provided $7 < x < 10$ for a fully correct method
		$x = 8$ and $y = 18.5$ oe		A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 4 | 4 marks

Family: Calculate and reverse discrete averages • Variant: Calculate the mean from a list or discrete frequency table

Solution

The median is 9:

$$(x+10)/(2)=9$$

$$x=8$$

The mean is 11, so the total is

$$6 \times 11=66$$

$$4+7+8+10+y+y=66$$

$$29+2y=66$$

$$2y=37$$

$$y=18.5$$

Answers: $x=8$, $y=18.5$.

- 2 The table shows information about the lengths of time, in minutes, 120 customers spent in a supermarket.

Length of time (L minutes)	Frequency
$20 < L \leq 30$	6
$30 < L \leq 40$	26
$40 < L \leq 50$	31
$50 < L \leq 60$	40
$60 < L \leq 70$	17

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean length of time spent by the 120 customers in the supermarket.

.....minutes
(4)

(Total for Question 2 is 5 marks)

Answer reference | May-Nov 2020 | Question 2

Family: Estimate grouped-data measures • Variant: Estimate a grouped total rather than the mean

2	a		$50 < L \leq 60$	1	B1	oe eg 50 - 60
	b	$25 \times 6 + 35 \times 26 + 45 \times 31 + 55 \times 40 + 65 \times 17$ $(150 + 910 + 1395 + 2200 + 1105)(= 5760)$			M2	For correct products using midpoints (allow one error) with intention to add. M1 for products using frequency and a consistent value within the range (allow one error) with intention to add or correct products using midpoints (allow one error) without addition
		"5760" ÷ "120"			M1	dep on M1
			48	4	A1	
						Total 5 marks

Worked solution

May-Nov 2020 | Question 2 | 5 marks

Family: Estimate grouped-data measures • Variant: Estimate a grouped total rather than the mean

Official final mark-scheme pages are embedded immediately after this question.

Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1 80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

Work out an estimate for the mean length of time the students spent dancing.

..... minutes

(Total for Question 1 is 4 marks)

Answer reference | January 2023 | Question 1

Family: Estimate grouped-data measures • Variant: Estimate a grouped total rather than the mean

Q	Working	Answer	Mark	Notes
1	$6 \times 11 + 18 \times 25 + 30 \times 23 + 42 \times 15 + 54 \times 6$ $(= 2160)$ or $66 + 450 + 690 + 630 + 324 (= 2160)$ [lower bound products are: 0, 300, 552, 540, 288] [upper bound products are: 132, 600, 828, 720, 360]		4	M2 for at least 4 correct products added (need not be evaluated) or If not M2 then award: M1 for consistent use of value within interval (including end points) for at least 4 products which must be added or correct midpoints used for at least 4 products and not added
	"2160" ÷ "80"			M1 dep on at least M1 Allow division by their Σf provided addition or total under column seen
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	27		A1
				Total 4 marks

Worked solution

January 2023 | Question 1 | 4 marks

Family: Estimate grouped-data measures • Variant: Estimate a grouped total rather than the mean

Official final mark-scheme pages are embedded immediately after this question.

- 15** Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

(Total for Question 15 is 3 marks)

Answer reference | January 2023 | Question 15

Family: Measure and compare spread • Variant: Find quartiles and IQR from discrete data

Question	Working	Answer	Mark	Notes
15			3	M1 4 and 34 clearly indicated – either in list or in working (condone 26 also indicated in list)
				A1 For IQR for team A = $34 - 4 (= 30)$
		The IQR for Team B was higher than the IQR for Team A oe or Team B had an interquartile range of “12” more than team A or The runs scored were more spread out for Team B than for Team A oe or The runs for Team A were more consistent oe		B1ft Must fit dep on IQR stated for team A Either comparing the IQR correctly or for giving a comparison in context about spread as long as not contradicted by further statements as this would be choice NOT Team B scored more runs than team A The average score of B is higher than the average score of A The IQR of A was 30 while the IQR of B was 42 The range of B was more than the range of A
				Total 3 marks

Worked solution

January 2023 | Question 15 | 3 marks

Family: Measure and compare spread • Variant: Find quartiles and IQR from discrete data

Official final mark-scheme pages are embedded immediately after this question.

- 1 The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

Worked solution

November 2025 | Question 1 | 4 marks

Family: Estimate grouped-data measures • Variant: Estimate a grouped total rather than the mean

Solution

Use midpoints 2.5, 7.5, 12.5, 17.5, 22.5.

$$2.5(8) + 7.5(7) + 12.5(3) + 17.5(10) + 22.5(2) = 330$$

$$\text{estimated mean} = (330)/(30) = 11$$

Answer: 11 km.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

$e = \dots\dots\dots$

$f = \dots\dots\dots$

$g = \dots\dots\dots$

(Total for Question 4 is 3 marks)

Worked solution

November 2025 | Question 4 | 3 marks

Family: Calculate and reverse discrete averages • Variant: Calculate the mean from a list or discrete frequency table

Solution

The range is 15:

$$17 - e = 15$$

$$e = 2$$

The mean is 9, so the total is

$$6 \times 9 = 54$$

$$2 + f + 7 + g + 12 + 17 = 54$$

$$f + g = 16$$

The median is 8.5:

$$(7 + g) / (2) = 8.5$$

$$g = 10$$

$$f = 16 - 10 = 6$$

Answers: $e=2$, $f=6$, $g=10$.

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)

Worked solution

November 2025 | Question 16 | 2 marks

Family: Present and interpret statistical diagrams | Variant: Recover totals or a diagram scale in reverse

METHOD / WORKING

Worked solution
November 2025 Question 16 2 marks
Family: Present and interpret statistical diagrams Variant: Recover totals or a diagram scale in reverse
Solution
The data are already in order:
1, 1, 3, 4, 6, 7, 8, 9, 10, 12, 15, 25, 30, 37, 50
There are 15 values.
$Q_1 = 4, Q_3 = 25$
$IQR = 25 - 4 = 21$
Answer: 21.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER